

Signal Manipulation: Shifting and Scaling in Time

```
import numpy as np
import matplotlib.pyplot as plt
```

```
signal = np.array([-5, -3, -1, 1, 3, 5, 7, 10, 17, 26])
origin = 4
time = np.arange(-origin, len(signal) - origin)
```

```
plt.figure(figsize=(10, 10))
```

```
plt.subplot(2, 2, 1)
plt.stem(time, signal, 'r+')
plt.title('Original Signal x(n)')
plt.xlabel('Time (n)')
plt.ylabel('Amplitude')
```

```
advance = 8
time_advance = np.arange(-origin - advance, len(signal) - origin - advance)
plt.subplot(2, 2, 2)
plt.stem(time_advance, signal, 'b+')
plt.title('x(n+8)')
plt.xlabel('Time (n)')
plt.ylabel('Amplitude')
```

```
scale_down = 1/16
time_scale_down = np.linspace(-origin * scale_down,
                               (len(signal) - origin) * scale_down,
                               len(signal))
plt.subplot(2, 2, 3)
plt.stem(time_scale_down, signal, 'g+')
plt.title('x(16n)')
plt.xlabel('Time (n)')
plt.ylabel('Amplitude')
```

```
scale_up = 10
time_scale_up = np.linspace(-origin * scale_up,
                             (len(signal) - origin) * scale_up,
                             len(signal))
plt.subplot(2, 2, 4)
plt.stem(time_scale_up, signal, 'm+')
plt.title('x(n/10)')
plt.xlabel('Time (n)')
plt.ylabel('Amplitude')
```

```
plt.tight_layout()
plt.show()
```

